Test 2 review day

CMSE 381

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Fri, April 18, 2025

Announcements

Announcements:

- midterm # 3 on Monday
- No office hour on Monday, Last office hour on Wednesday.
- Final Project submission on D2L. Due April 25.
- Student Perception of Learning Survey (SPLS): Please fill out this

Based on the questions submitted

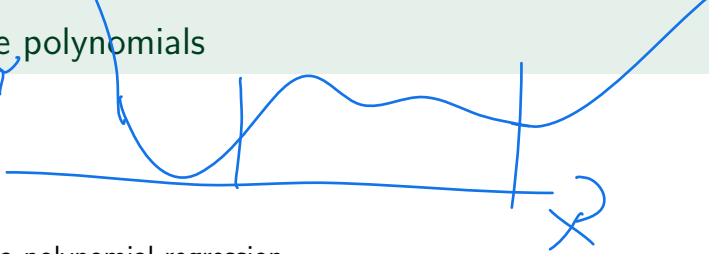
linear linear or nonlinear

- Cubic spline
- Difference between SVC and SVM
- Neural network stuff

Section 1

Cubic Splines

Piecewise polynomials

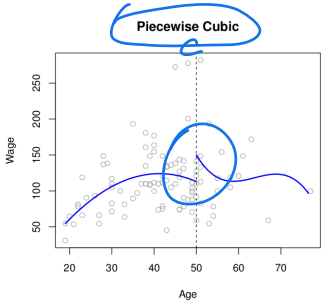


- Fit a polynomial regression

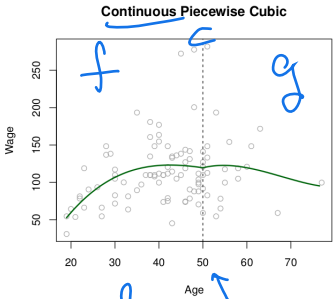
$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$

- Let the β_i's be different at different locations of the range.

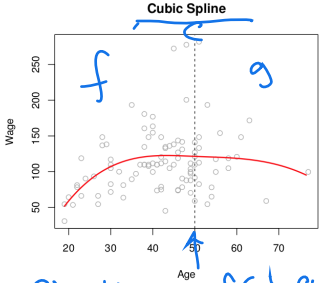
Building up to cubic splines



no constraint at knot



$f(c) = g(c)$



① continuous. $f(c) = g(c)$

② cont 1st deri $f'(c) = g'(c)$

③ conti 2nd deri $f''(c) = g''(c)$

$$y_i = \begin{cases} \beta_{01} + \beta_{11} x_i + \beta_{21} x_i^2 + \beta_{31} x_i^3 + \epsilon_i & \text{if } x_i < c \\ \beta_{02} + \beta_{12} x_i + \beta_{22} x_i^2 + \beta_{32} x_i^3 + \epsilon_i & \text{if } x_i \geq c. \end{cases}$$

Test your understanding: PollEv

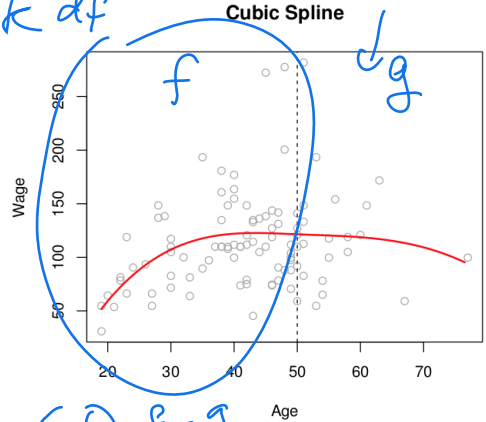
Cubic splines: degrees of freedom

in general: k knots $\rightarrow$ $4+k$ df

$$f(x) = \begin{cases} \beta_0^1 + \beta_1^1 x + \beta_2^1 x^2 + \beta_3^1 x^3 & x < c \\ \beta_0^2 + \beta_1^2 x + \beta_2^2 x^2 + \beta_3^2 x^3 & x > c \end{cases}$$

no constraints: 8 par

Cubic spline: 5 df

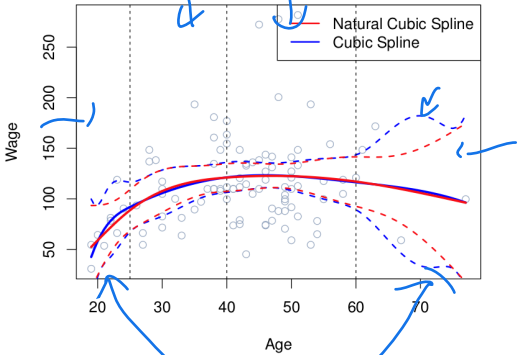


① $f = g$
② $f' = g'$
③ $f'' = g''$

Notes on cubic splines

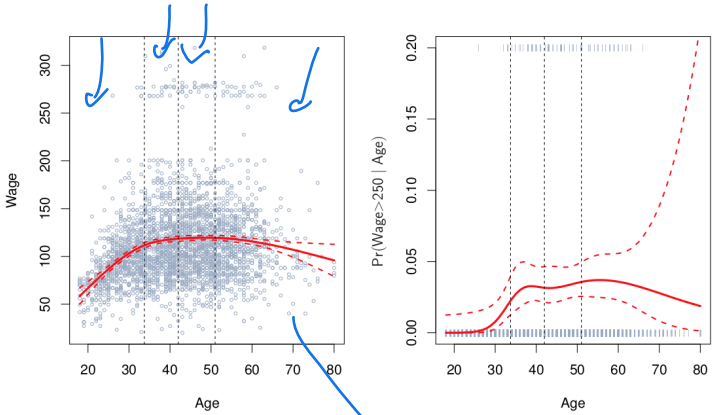
middle → good

- fit a smooth curve
- natural cubic spline.
  boundary. linear.



boundary: high variance

Where to put the knots?



① evenly split over X

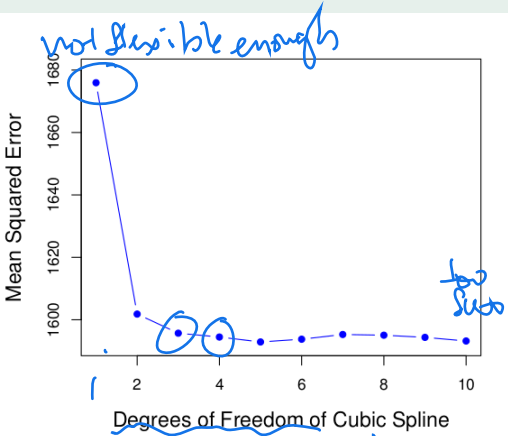
② by percentile: 25% 50% 75%

How many knots to use?

When in doubt, Cross-Validate.

bias-variance.

- # knots control flexibility.
- CV (k-fold)



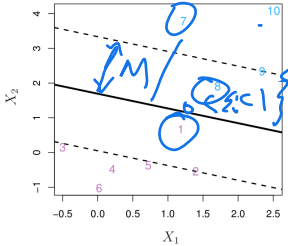
Section 2

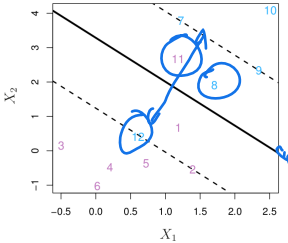
SVC vs SVM

linear

can do everything

Support Vector Classifier





$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

distance to the hyperplane from the correct side

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i)$$

$\geq M$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

total budget of margin violation

$\epsilon_i = 0$ if point is on the correct side of the margin

So many variables

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

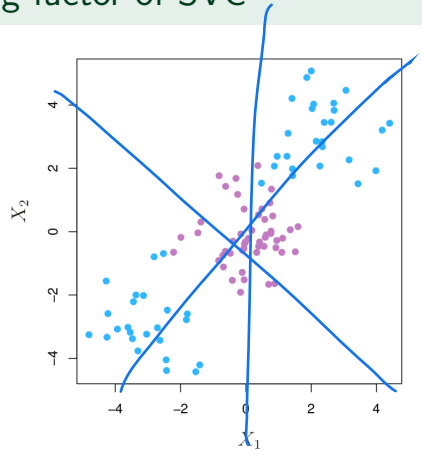
$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

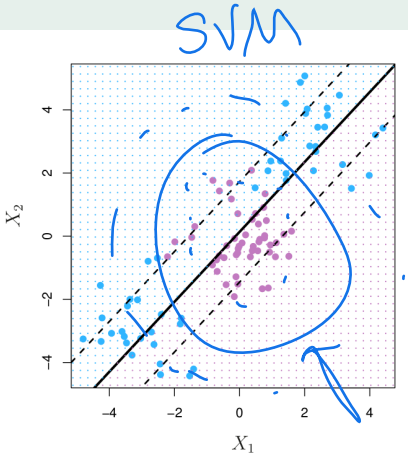
$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

- C is nonnegative tuning parameter
- M is the width of the margin
- $\varepsilon_1, \cdots, \varepsilon_n$ are slack variables allowing observations to go to the other side

Limiting factor of SVC





More general way of thinking about constraints – Kernels

$$y_i f(x_i) \geq \boxed{M(1-\epsilon_i)}$$

$f(x_i)$ should give you some more general way of talking about how far is point x_i away from the decision boundary.

distance of point to hyperplane

Some kind of boundary

(on correct side)

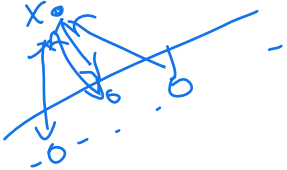
The kernel

$f(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots$

$$\boxed{f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i K(x, x_i)}$$

$$K(x_i, x_i')$$

Test your understanding: PollEv



$K(x, x_i) = \langle x, x_i \rangle$

inner product

$\rightarrow$ equivalent to SVC

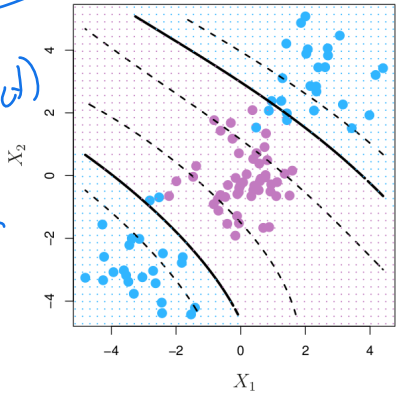
i.e. it is linear.

A polynomial kernel

$$K(x_i, x_{i'}) = \left(1 + \sum_{j=1}^{p} x_{ij} x_{i'j}\right)^d$$

• more flexible than linear.
(inner product)

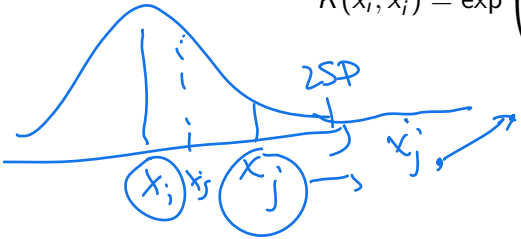
3 degree.

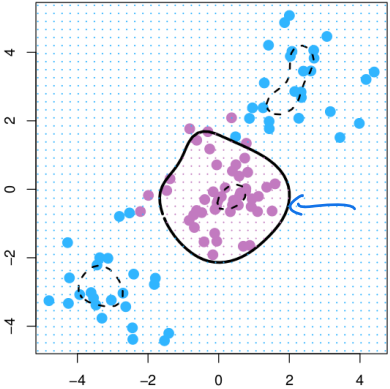


A radial kernel

Gaussian Kernel

$$K(x_i, x_i') = \exp\left(-\gamma \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2\right)$$





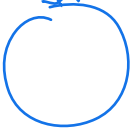
TL;DR

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i K(x, x_i)$$

point

support vector

$f(x)$



Kernels

- Linear

$$K(x_i, x_{i'}) = \sum_{j=1}^{p} x_{ij} x_{i'j}$$

SVC

- Polynomial

$$K(x_i, x_{i'}) = \left(1 + \sum_{j=1}^{p} x_{ij} x_{i'j}\right)^d$$

global nonlinear

- Radial

local nonlinear

$$K(x_i, x_i') = \exp\left(-\gamma \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2\right)$$

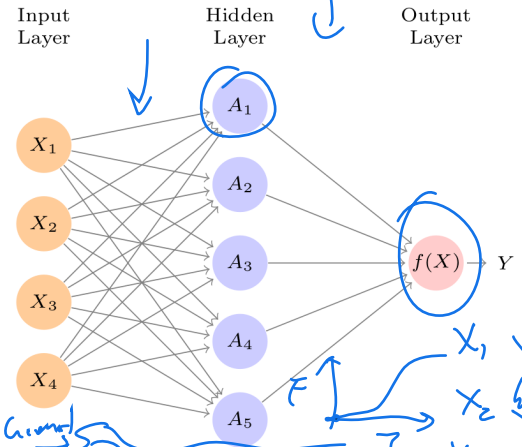
Gaussian kernel.

Section 3

Neural Nets

Feed Forward Neural Network: The cartoon

input-output
Transfer function



activation.

weights

$$A_k = h_k(X) = g(w_{k0} + \sum_{j=1}^{p} w_{kj} X_j),$$

$$f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k$$

linear or Softmax if classification.

A_1

activation function

$X_1 \xrightarrow{w_{11}}$

$X_2 \xrightarrow{w_{12}}$

$X_3 \xrightarrow{w_{13}}$

$X_4 \xrightarrow{w_{14}}$

$Z_1 = w_{10} + \sum_j w_{1j} X_j \xrightarrow{g(\,)} g(Z_1) \rightarrow A_1$

bias

nonlinearity

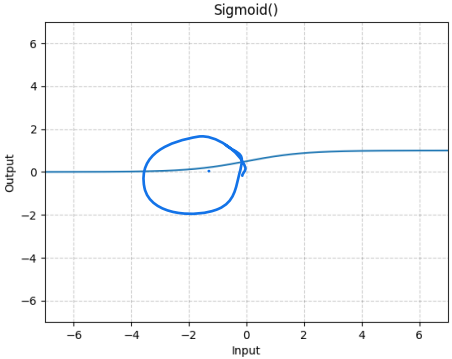
relu

sigmoid

Choices for activation function

Sigmoid:

$$g(z) = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}}$$

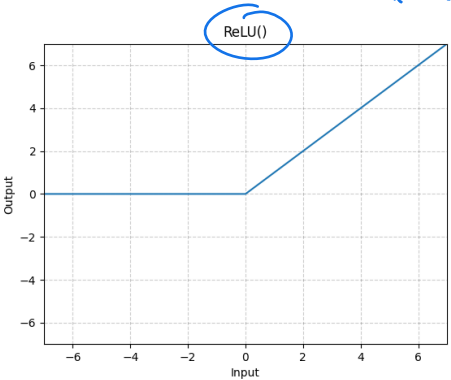


much easier to compute.

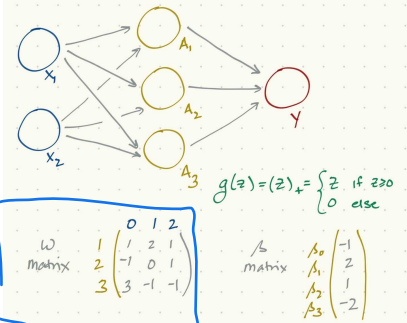
ReLU: Rectified linear unit

$$g(z) = (z)_+ = \begin{cases} 0 & \text{if } z < 0 \\ z & \text{else.} \end{cases}$$

max(0, z)



Matrix version



$$A_k = h_k(X) = g(w_{k0} + \textstyle\sum_{j=1}^{p} w_{kj} X_j),$$

$$A = g(\mathbf{W}\ \mathbf{X}) \qquad \mathbf{X}^T = (1\ X_1\ X_2\ \cdots\ X_p)$$

$$f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k$$

$$Y = \beta \cdot \mathbf{A} \qquad \mathbf{A}^T = (1\ A_1\ A_2\ \cdots\ A_K)$$

$(\beta_0\ \beta_1\ \beta_2\ \cdots\ \beta_K)$